

Technical Exam - Task

Goal: Build a small end-to-end data pipeline that produces a fine-tuning-ready dataset for a media bias classifier. We care more about your **decisions, judgment, and code quality** than about achieving state-of-the-art results.

Time budget: ~4-6 hours of focused work. If you reach the time limit, stop and document what you'd do with more time. That's a valid and valued outcome.

Deliverable: A git repository (or zip) containing your code, the dataset you produced, and a README.md report.

What We're Evaluating

You will be assessed on:

1. **Judgment and scoping:** Did you make sensible choices given the time budget? Did you recognize and document trade-offs?
2. **Code quality:** Is the code readable, organized, and something you could hand to a teammate?
3. **Data quality awareness:** Do you understand the limitations of what you produced?
4. **Communication:** Is the report clear, honest, and useful?
5. **Technical execution:** Does the pipeline run end-to-end and produce sensible output?

We are **not** evaluating final classifier accuracy. We are not running your code against a hidden test set. This is about how you think and work.

We allow/expect you to use AI coding assistants. But: expect to be asked about specific parts of your code and the choices behind them. Submissions that look like AI output the candidate can't explain will not pass.

- If possible, include your prompts, and if you used an agent, the plan it produced.

The Task

1. Choose a bias detection sub-task

Pick one specific sub-task of "media bias detection":

- selected from a published media bias survey
- Your sub-task should be measurable in both text and image content
- Justify your choice in the report: why this sub-task, why it's tractable in the time budget, and why it's measurable from your data.

2. **Collect articles:**

Build a small crawler that fetches articles

- Target (100-300 articles)
- Libraries can be used

3. **Processing:**

Clean and structure the articles into a format suitable for downstream labeling. Decisions to make and document briefly in report.

- e.g., (verify content/clean, split in sentences, images)
- Schema of your processed data

4. **Pre-Label:**

Find an off-the-shelf classifier on HuggingFace whose task is related but *not identical* to your chosen sub-task (e.g., a subjectivity classifier as a proxy for opinion-vs-fact framing). Run it on your data and store predictions in your schema.

- In your report, explain why this model is a reasonable proxy and where you expect it to disagree with your target task.

5. **LLM + VLM annotation:**

Annotate a sample (50-100 samples) with both an LLM and a VLM. Include the prompts in the repository. You can use a small local model.

- LLM: produce a label for your chosen task plus a short natural-language explanation.
- VLM: for each article's main image, produce a description and an assessment of whether the image (and related text) contains bias, plus a short explanation.

6. **EDA**

- Descriptive analysis of your dataset.
- Agreement analysis between the proxy classifier labels (step 4) and the LLM annotations (step 5)

7. **Report**

- Write a short report on the decisions made (can be README.md)
- Required:
 - Task chosen and why
 - Pipeline overview (one paragraph or a small diagram)
 - Decisions made, including:
 - Why your chosen proxy classifier is a reasonable fit
 - Shortcuts taken (e.g., for time reasons)
 - Risks of the approach
 - AI usage